

Homestay Phu Nhuan

Automated Booking Platform • AI Chatbot • Self Check-in

An automated homestay booking platform that lets guests check availability, book a time slot, pay online and receive self check-in credentials with minimal owner intervention.



Core value

Guests can book at night, complete payment and receive check-in information immediately — allowing the owner to avoid monitoring calls or messages overnight.

ROLE

Product Designer & Full-stack Web Developer

OWNERSHIP

Built end-to-end by me

DURATION

Jun-Jul 2026

PROJECT TYPE

Production web application for real homestay operations

SCOPE

Customer website • booking workflow • serverless APIs • Firebase data model • third-party integrations

PLATFORM

Responsive web application

20+

weekday users/day

avg

50+

Fri-Sun users/day

avg

180

monthly bookings

5 HOMES

40%

monthly occupancy

last month



Business problem

From manual owner-managed booking to a 24/7 self-service workflow.



Common manual model

- 1 Guests message/call to ask if a room is available.
- 2 Owner checks Excel or manual records.
- 3 Late-night demand is easy to miss if the owner is asleep.
- 4 Payment confirmation and access-code delivery are manual.
- 5 Higher risk of schedule mismatch or missing information.



Automated solution

- 1 Guests check realtime availability by themselves.
- 2 System validates if the requested slot can be booked.
- 3 Guests book, pay and receive check-in information online.
- 4 Late-night booking can still finish without owner intervention.
- 5 Booking/support requests are pushed to a Telegram group.



Core operational benefit: the owner does not need to monitor calls overnight, while the system can still sell rooms and notify the team.

Realtime Data & Availability



The website and chatbot do not answer from static content. Each check reads the latest data from the system.

What is realtime?

- ✓ Availability and booked slots for each HOME
- ✓ Pricing, time packages, surcharges and discounts
- ✓ HOME information, amenities and room photos
- ✓ Temporary room locks while QR payment is pending
- ✓ Booking and payment status

Firebase Realtime Database

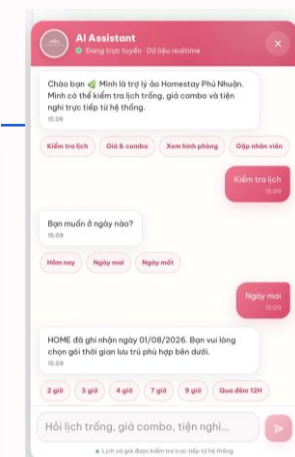
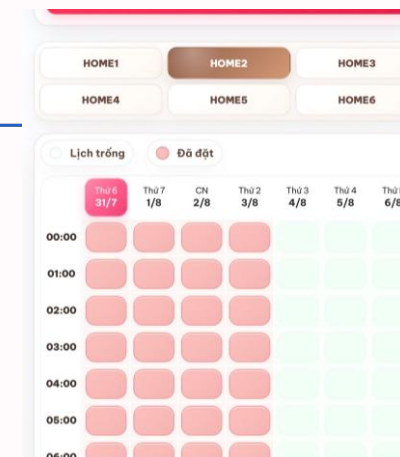
Operational system of record

DB

Website

Chatbot

Admin



Gemini does not guess availability or prices: business data is read from Firebase, while AI only helps phrase the response more naturally.

Automated Booking Journey

From room browsing to access codes: the guest moves through a clear self-service flow.



UI

HOME1

Ngày: 1 Aug 2026

Gói: Qua đêm 12H · 500.000đ

Giờ bắt đầu: 00:00 - 23:30 - mỗi 30 phút

15:30

Tổng tiền thanh toán: 500.000đ

Thanh toán toàn bộ bằng QR payOS

Còn trống, có thể giữ phòng

Select HOME, date, package and time

PAY

2 người

500.000đ

Tổng tiền thanh toán: 500.000đ

Thời gian hoàn tất: 15 phút

Bạn thanh toán toàn bộ số tiền trên qua QR payOS. Booking được xác nhận sau khi hệ thống ghi nhận thanh toán thành công.

Tên khách: Nhập tên của bạn

Số điện thoại: Ví dụ: 0902932808

Confirm details and total price

HOLD

500.000đ

500.000đ

Thời gian hoàn tất: 15 phút

Bạn thanh toán toàn bộ số tiền trên qua QR payOS. Booking được xác nhận sau khi hệ thống ghi nhận thanh toán thành công.

Dang chờ thanh toán

Bạn chưa cung cấp ảnh giấy tờ. Mục này không bắt buộc.

Thời gian giữ phòng còn lại: 14:10

15-minute hold while paying

KEY

Thông tin check-in và checkout

Mã cửa được tạo riêng cho booking này. Chỉ chia sẻ cho người cùng phòng.

Mã check-in: 584500

Khóa ngoài - Dùng để vào

HIỆU LỰC: 14:00 31/07/2026 → 16:00 31/07/2026

Mã checkout: Khóa trong - Dùng để ra

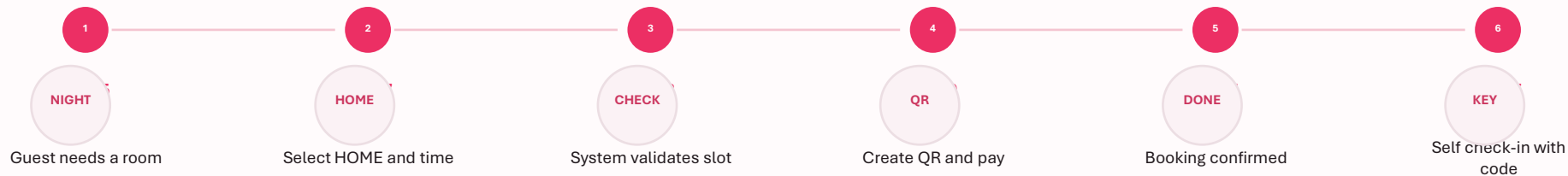
Worker calls TTLock API to issue time-bound codes

After successful payment, a Cloudflare Worker calls TTLock API to issue time-bound check-in/checkout passcodes.

Outcome: booking, payment and check-in can be completed even when the owner is not online.

Night Booking, Smart Lock & Telegram

Built for urgent, after-hours demand while reducing dependency on the owner.



Time-bound access control

- Check-in code is valid only from the booking check-in time.
- Checkout code follows the booking window; exceptions require owner/admin support.
- Cloudflare Worker calls TTLock API to issue/update passcodes within the booking window.

Operational meaning

No back-and-forth confirmation is required; the website acts as a 24/7 digital front desk.

Kiểm tra booking và thanh toán

Kiểm tra lại lịch, tổng tiền, nhập thông tin liên hệ rồi tạo QR payOS. Booking được xác nhận sau khi thanh toán thành công.

Số người	2 người
Giá gói (tối đa 2 người)	500.000đ
Tổng tiền thanh toán	500.000đ
Thời gian hoàn tất	15 phút

Bạn thanh toán toàn bộ số tiền trên qua QR payOS. Booking được xác nhận sau khi hệ thống ghi nhận thanh toán thành công.

Đang chờ thanh toán

Bạn chưa cung cấp ảnh giấy tờ. Mục này không bắt buộc.

Thời gian giữ phòng còn lại 14:10

Mã đơn payOS

548516619240

Check-in	14:00 31/07/2026
Check-out	16:00 31/07/2026
Đã thanh toán	2.000đ

MÃ CỬA BẢO MẬT

Thông tin check-in và checkout

Mã cửa được tạo riêng cho booking này. Chỉ chia sẻ cho người cùng phòng.

Mã check-in
Khóa ngoài - Dùng để vào

584500

HIỆU LỰC:
14:00 31/07/2026 → 16:00 31/07/2026

Mã checkout
Khóa trong - Dùng để ra

574163

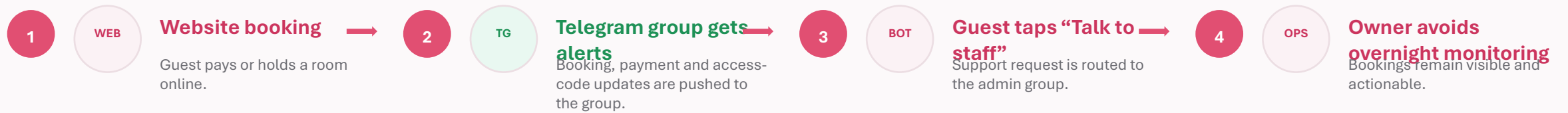
HIỆU LỰC:
14:00 31/07/2026 → 16:00 31/07/2026

Mã check-in dành cho khóa ngoài và có thể bắt đầu trước

Telegram Bot & Support Automation



No hotline monitoring required, while bookings and support requests remain visible to the team.



TG

Operational Telegram group

New website booking

HOME2 • Overnight 12H • 500,000 VND

Guest paid through PayOS

Access codes created • valid within booking window

Guest needs staff support • admin follows up

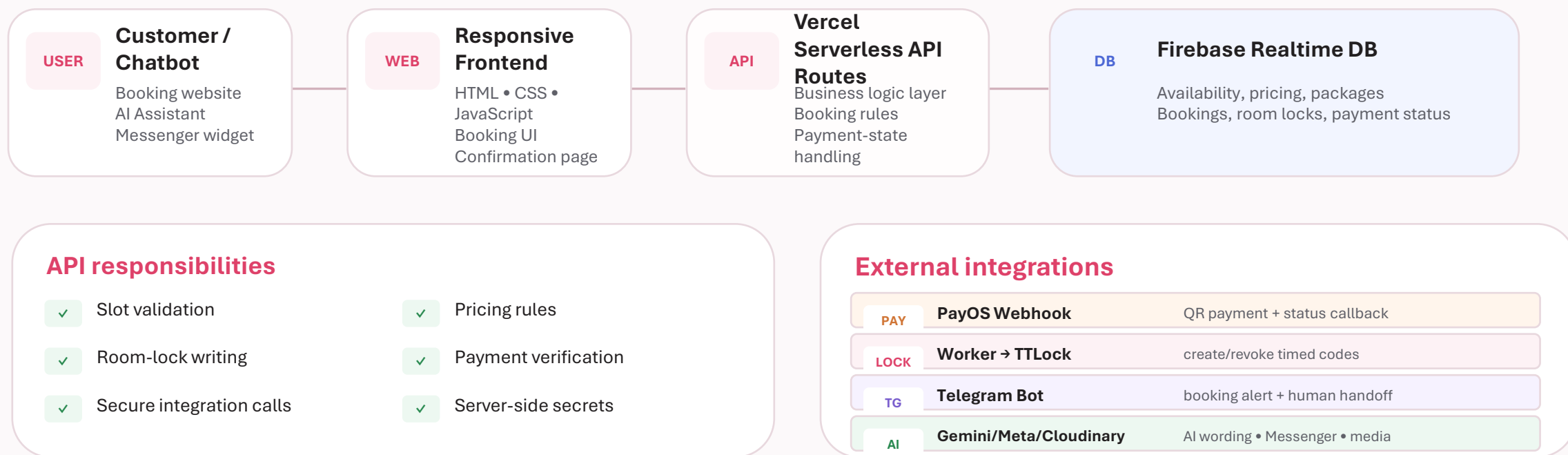
OK

Operational point

The website can sell rooms 24/7;
Telegram ensures owner/admin
visibility without continuous phone
monitoring.

System architecture diagram

The API layer orchestrates requests, reads realtime data, validates business rules, verifies payment state, calls Worker/TTLock and sends operational notifications.



! Core rule: the frontend does not decide availability or door access. APIs verify business data, payment state and booking rules before writing Firebase or calling the Worker / integrations.

Webhooks & Integration Flow

Separates internal data, payment, smart-lock access and operational notifications for real production usage.

PAY

PayOS webhook / callback

PayOS sends transaction status to the API. Booking is confirmed only after valid payment status.

CF

Cloudflare Worker → TTLock API

The Worker keeps lock secrets and calls TTLock to create / revoke time-bound codes.

TG

Telegram Bot notification

Bookings, payment, access codes and support requests are pushed to the operations group.

MSG

Meta Messenger webhook

Receives messages and verifies Meta requests using HMAC signature validation.

AI

Gemini grounded response

The system reads Firebase first; Gemini only phrases the answer naturally.

IMG

Cloudinary upload

Room photos and stay documents are stored in Cloudinary, separate from operational data.



Production-ready boundary

- The frontend only calls controlled APIs; secrets stay server-side / Worker-side.
- PayOS confirms payment; the Worker calls TTLock API to create time-bound access codes.
- Telegram, Cloudinary and Gemini have clear responsibilities.

Tech Stack, Reliability & Security

Production implementation: frontend, API, realtime data, integrations, security, reliability and QA.

WEB

Frontend & API

- ✓ Responsive HTML/CSS/JS UI
- ✓ Vercel Serverless API Routes
- ✓ Firebase Admin SDK server-side
- ✓ Luxon timezone handling

DB

Data & state

- ✓ Firebase Realtime Database
- ✓ HOME info / pricing / packages
- ✓ Bookings, payments, room locks
- ✓ Temporary hold + payment status

INT

Integrations

- ✓ PayOS webhook / callback
- ✓ Cloudflare Worker + TTLock API
- ✓ Telegram Bot notifications
- ✓ Cloudinary, Gemini, Meta Messenger

SEC

Security & QA

- ✓ Firebase ID token verification
- ✓ Role-based admin authorization
- ✓ Meta webhook HMAC check
- ✓ Automated tests + health checks

✓

Reliability safeguards

- ✓ Booking conflict validation
- ✓ 15-minute room lock
- ✓ Payment verified first
- ✓ Time-bound TTLock codes
- ✓ Telegram visibility

QA

QA & technical hardening

Tests cover availability, pricing, payment, auth, webhook and owner notification. Next: idempotent callbacks, retry/logging, recovery flow and funnel analytics.

CF

Cloudflare Worker keeps lock secrets and calls the TTLock API to create / revoke time-bound access codes.

Operational value comparison against the common homestay model

Key difference: automation across booking, payment, access codes and guest support.

01



End-to-end automation

Availability, pricing, payment and access codes run through one digital flow.

02



24/7 operations

Guests can book at night, even when the owner is offline.

03



Less manual work

Fewer manual replies for availability, pricing, transfers and access codes.

04



More private experience

Guests self-select time, pay online and self check-in/out.



Most important: the owner can rest while the system receives bookings, processes payment and supports guests by process.



Common model in Ho Chi Minh City vs Homestay Phu Nhuan

	Common model	Homestay Phu Nhuan
Check availability	Manual / inbox inquiry	Realtime self-check
Validate slot	Depends on owner	Automatic slot validation
Hold room during payment	Manual confirmation	15-minute room lock
Send access code	Sent manually after closing	Auto-generated timed codes
Night booking	Easy to miss if owner sleeps	Book & pay 24/7
Talk to staff	Handled separately	Forwarded to Telegram group

Results & Business Value

Operational metrics are reported as observed; conversion rate is not claimed without full funnel tracking.

**20+**

weekday users/day
weekday traffic

**50+**

users/day Fri-Sun
avg Fri-Sun

**180**

monthly bookings
last month • 5 HOMEs



Business value delivered

- ✓ Guests check availability, prices and HOME information before booking.
- ✓ Reduces manual schedule checks, pricing replies, payment confirmation and access-code delivery.
- ✓ PayOS + room lock makes room-hold status clearer.
- ✓ Telegram keeps owner/admin informed of bookings and support requests.

**5**

HOMEs
managed in system

**40%**

monthly occupancy
last month

**10**

PayOS bookings
first 15 days



PayOS adoption note

10 PayOS bookings during the first 15 days; remaining bookings still came from Facebook/Zalo.



Conversion rate not claimed

Requires funnel tracking: visitor → check availability → start booking → create QR → paid booking.



Measured results are separated from expected benefits to keep the portfolio honest, clear and credible.

Role, Scope & Key Learnings

A clear summary of ownership, scope, technical learning outcomes and demo access.



My role

- ✓ Product Designer & Full-stack Web Developer
- ✓ Built end-to-end independently
- ✓ Analyzed the operational workflow
- ✓ Designed UX and responsive UI
- ✓ Built frontend, APIs and Firebase data model
- ✓ Integrated PayOS, TTLock, Telegram, Gemini and Cloudinary



Project scope

- ✓ Customer website & chatbot widget
- ✓ Realtime schedule / pricing / HOME data
- ✓ Booking flow + PayOS payment
- ✓ Time-bound TTLock credentials
- ✓ Telegram booking alert & human handoff
- ✓ Admin operations with data privacy



Key learnings

- Separate business data from AI wording to avoid incorrect pricing/availability.
- Booking systems need room locks and payment-state validation to reduce conflicts.
- Third-party integrations require clear error states, logging and recovery paths.



Live demo: homestayphunhuan.vercel.app



Contact: see CV / LinkedIn